OMNI CAMERA ROVER | REV C PROTOTYPE

1. Controller

File: controller.kicad_sch

2. Motors

File: motors.kicad_sch

3. Charging

File: charging.kicad_sch

4. Regulator

File: regulator.kicad_sch

5. Interlock

File: interlock.kicad_sch

Battery protection

File: battery_protection.kicad_sch

PWR_FLAG

PWR_FLAG

PWR_FLAG

PWR_FLAG

PWR_FLAG

PWR_FLAG

Editable functional sheets. Global net names link sheets.
Prototype: electrical, thermal and mechanical qualification remains open.

Flex camera

File: camera.kicad_sch

Cell undervoltage load shutdown

File: cell_undervoltage.kicad_sch

Cell temperature charging window

File: charge_temperature.kicad_sch

Nathaniel Sanford	
Sheets: /	
File: omni_ugv.kicad_sch	
Title: Omni UGV / Rev D	
Size: A3	Date: 2026-11-06
KiCad E.D.A. 10.0.8	Rev: D — prototype
	Id: 1/10

Controller, camera and service points

RESET / enable RC

R1
10k

C1
1u/16V

SW1
RESET

U1
ESP32-S3-WROOM-1-N8R8

3V3

EN

IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16

TXD0
RXD0
IO19
IO20
IO26
IO33
IO34
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48

PSRAM

GND

C2
10u/16V

C3
100n/16V

Waveshare OV2640 | documented 2 x 9 interface

Status indicator

R3
1k

D1
STATUS GREEN

BOOT strap

R2
10k

SW2
BOOT

[SPEC] GPIO35 / 36 / 37 reserved for N8R8 PSRAM.

Camera reset RC

R4
10k

C4
100n/16V

C5
10u/16V

Service points (named nets continue on other sheets)

TP1
GND

TP2
+3V3

TP3
VBAT

TP4
VMOTOR

TP5
SYS_5V

TP6
ESTOP

TP7
LED_STATUS

TP8
DRIVE_ENABLE

TP9
FAULT_I12

TP10
FAULT_I13

TP11
RESERVED

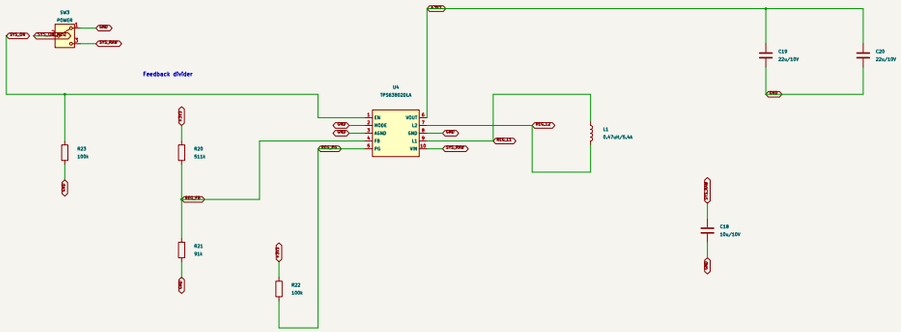
Nate | Omni camera rover	
Sheets: /1. Controller/	
File: controller.kicad_sch	
Title: Controller, camera and service points	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Rev: D — prototype
	Id: 2/10

3.3 V buck-boost supply

System enable

Feedback divider

Output capacitors



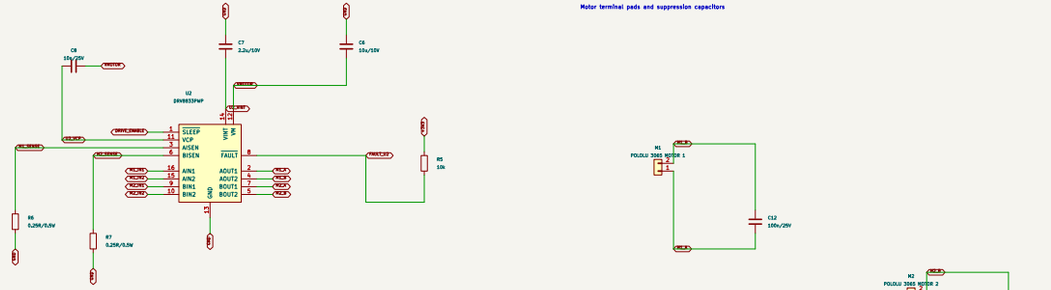
[TARGET] Keep switching loops short. Place input/output capacitors at U1.
[TBD-MEASURE] Verify startup, load transients and capacitor effective capacitance.

Nate | Omni camera rover		
Sheet: /4. Regulator/		
File: regulator.kicad_sch		
Title: 3.3 V buck-boost supply		
Size: A3	Date:	Rev: D — prototype
KiCad E.D.A. 10.0.8		Id: 3/10

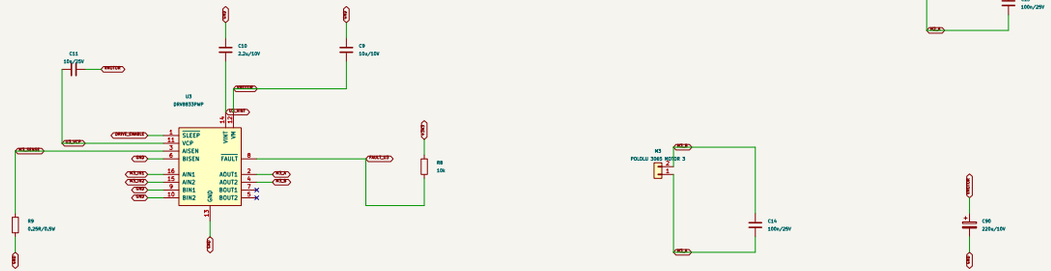
Three independent motor channels

U2: wheels 1 and 2

Motor terminal pads and suppression capacitors



U3: wheel 3: unused bridge pins retained as designed



[TARGET] Route motor current away from logic returns.
[TBD-MEASURE] Braking overshoot, driver temperature and carpet load.

Nate | Omni camera rover		
Sheet: /2. Motors/		
File: motors.kicad_sch		
Title: Three independent motor channels		
Size: A3	Date:	Rev: D — prototype
KiCad E.D.A. 10.0.8		Id: 4/10

USB motor interlock and battery measurement

USB presence clamps DRIVE_ENABLE

Back-to-back motor supply switch

Switched battery measurement

[TARGET] Reset / absent commands keep motors disabled.
[TBD-MEASURE] Verify USB insertion/removal and motor braking transients.

Nate | Omni camera rover	
Sheets: /5. Interlock/	
Files: Interlock.kicad_sch	
Title: USB motor interlock and battery measurement	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Rev: D — prototype
	Id: 5/10

USB input, charging and cell connections

USB data to ESP32 native USB

Raw cell -> protection sheet

Charge / system power path

Cell-bonded charger probe

[SPEC] J2 CELL_N is not system GND: R36 is the only intended return path.
J4: independent insulated two-wire 103AT-2 probe on the cell.
Motor current uses protected VBAT, not charger SYS_RAW output.

Nate | Omni camera rover	
Sheets: /3. Charging/	
Files: charging.kicad_sch	
Title: USB input, charging and cell connections	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Rev: D — prototype
	Id: 6/10

1S protection and cell-temperature monitoring

Cell voltage / current sensing

R37
100R 1%
C24
100n/16V X7R
R38
20R 1%
U7
BQ296212TRUGR
BAT
VDD
VSS
CS
CTR
PACK
DSG
CHG
VCELLP
VBAT
VCELLN
PROT_PACK
PROT_DSG
PROT_CHG

R39
1k 1%
R36
10m 1% 1W

Common-drain high-side protection FET pair

Q7
CSD17306Q5
Q8
CSD17306Q5
S_[1-3]
D_[5-9]
RA1
10M
RA0
10M

TH2 -> J5 -> GPIO1 / ADC1

J5
CELL NTC — 10KNE
RA2
10k 1%
RA3
1k 1%
TH2
103AT-2 CELL-MOUNTED
C25
100n/16V X7R
CELL_TEMP_ADC

TH1 -> J4 analog charge window

TH1
103AT-2 CELL-MOUNTED

#FLG20
PROTECTION_PWR_FLAG
#FLG21
PROTECTION_PWR_FLAG

[SPEC] TH1 and TH2 are OFF-BOARD, bonded to the cell through electrical insulation.
Separate two-wire probes: do not connect either sensor to new CELL_N.
[TARGET] Firmware must stop driving on overtemperature, open/short probe or state readings.

Nate | Omni camera rover	
Sheet: /Battery protection/	
File: battery_protection.kicad_sch	
Title: 1S protection and cell-temperature monitoring	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Rev: D — prototype
	Id: 7/10

J3
M0031 CAMERA FPC24

U8
TLV75533PDBVR
U9
TLV75528PDBVR
C26
2.2µ/10V X7R
C27
2.2µ/10V X7R
C28
2.2µ/10V X7R
C29
2.2µ/10V X7R
R44
4.7k
R45
4.7k

M0031 only. Verify flex contact side, thickness and pin-1 before assembly.
1.3 V core / 2.8 V analog / 3.3 V I-O. No interchangeable generic OV2640 pinout.

Sheet: /Flex camera/	
File: camera.kicad_sch	
Title: OV2640 M0031 flex interface	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Rev: D — prototype
	Id: 8/10

Independent cell undervoltage shutdown

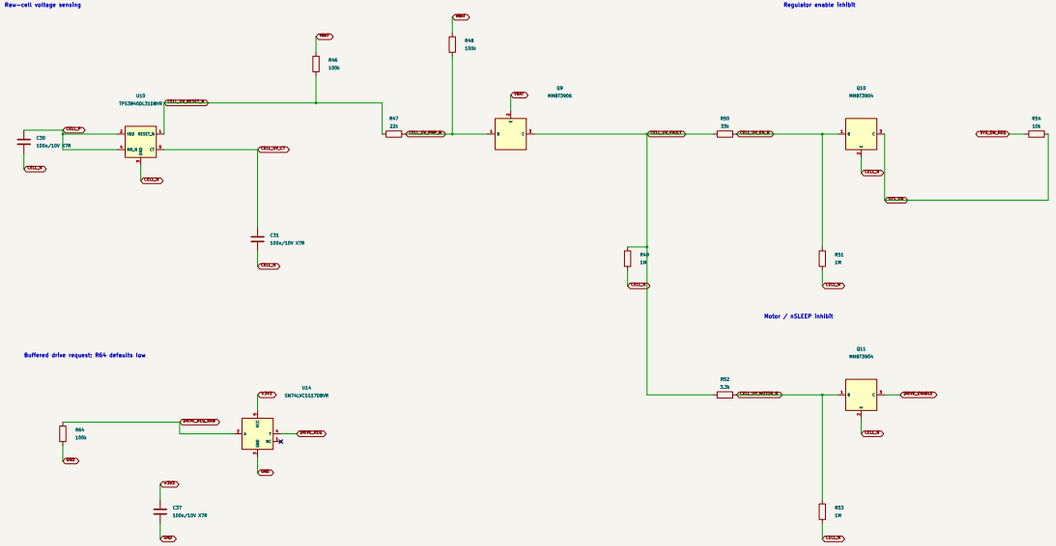
Raw-cell voltage sensing

Protected-rail inverter

Regulator enable inhibit

Motor / nSLEEP inhibit

Buffered drive request: R64 defaults low



3.1 V nominal falling cutoff; approximately 3.3 V recovery, CT delays release, not fault assertion.
Q9 and pullups use protected VBAT; only the nanopower supervisor remains on raw CELL_P after U7 cutoff.
Charging is independent of SYS_ON, so USB can recover a discharged cell while ESP32/camera stay off.
U7 retains its factory 2.6 V emergency cutoff. Firmware should stop normally before this hardware trip.

Nate | Omni camera rover	
Sheet: /Cell undervoltage load shutdown/	
File: cell_undervoltage.kicad_sch	
Title: Independent cell undervoltage shutdown	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Rev: D — prototype Id: 9/10

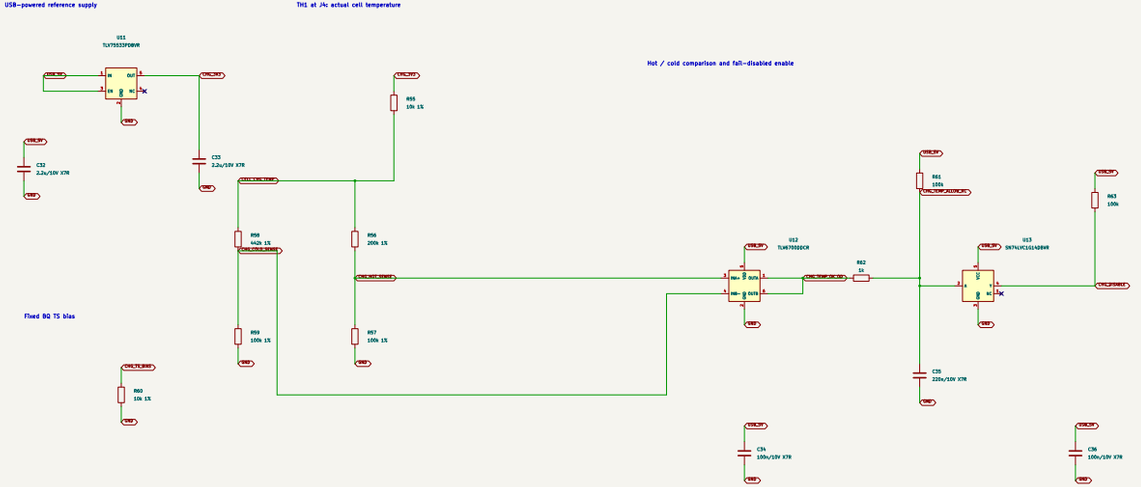
Independent cell temperature charging window

USB-powered reference supply

TH1 at Jc: actual cell temperature

Hot / cold comparison and fail-disabled enable

Fixed BQ TS bias



TH1 is an electrically insulated OFF-BOARD 103AT-2, bonded to the cell. Open/short probe disables charging.
U12 open-drain outputs both release only within the temperature window. U13 then drives CE low.
R61/C35 delay charge enable at USB startup; R62 limits capacitor discharge into U12. No ESP32 dependency.
R60 intentionally replaces the direct TS thermistor. This independent circuit now decides cell charge temperature.
Tolerance calculation uses the NTC beta model; verify actual resistance-temperature curve and trip limits on a fixture.

Nate | Omni camera rover	
Sheet: /Cell temperature charging window/	
File: charge_temperature.kicad_sch	
Title: Independent cell temperature charging window	
Size: A3	Date:
KiCad E.D.A. 10.0.8	Rev: D — prototype Id: 10/10